

Leveling Judge Stringency: What Survives a Two-Bed Fallback-Rescue Format Artifact?

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Abstract

Answer parsers are part of an evaluation’s measurement instrument: a permissive fallback can recover a correct value even when the requested format is absent. We audit this effect on two frozen beds by separating format coverage, accuracy conditional on explicit format, and their strict joint product. Both beds use free-form generative GSM8K; because the frozen artifacts define no multiple-choice wrapper, option-count and chance-rate baselines do not apply, and cross-bed levels describe selected populations rather than capability. Across all Bed 1 arms, the format channel is 5.8–9.5 pp higher, which is far too small to explain the 35–67× rescue-ratio contrast. Order is rule-dependent: the raw point-estimate rule gives 3/28, whereas the favorable paired gate—pre-registered in estimand wording but operationalized post hoc—gives 0/23 and applies only to these two beds. The two F_{strip} counts use different calibers, and the separately reported adversarial bounds use different denominators; these quantities are not interchangeable (Table 6). Finally, the registered pattern question and its primary conditional MDA80 are not identified from the shipped summaries: the available data cannot decide.¹

1 Introduction

Evaluation code is part of the measurement instrument. When a parser first looks for an explicit answer format and then rescues an otherwise unparseable response from its final position, the reported accuracy mixes at least two events: producing the requested format and answering correctly within that format. A comparison is especially fragile when different experimental beds invoke the rescue path at very different rates.

We study this failure mode in two frozen beds evaluated by canonical v1.1. Table 1 makes the central contrast reconstructible: the rescue share differs by 35–67× between B1-32 and the three Bed 2 arms used in the primary level/pattern analyses. This prevents a raw canonical level from carrying the same operational meaning across beds, but does not imply that every relation measured by the judge disappears.

This paper separates three questions that are often conflated:

1. Level: can normalized accuracies be read as a cross-bed capability ordering?
2. Order: does removing fallback ever reverse an observed strict ordering between arms?
3. Pattern: does a registered canonical-tie/strict-separation label revert to a tie after judge normalization?

Our contribution is a measurement audit with pre-registered estimands and primary-arm assignment; the favorable order gate and final synthesis were pre-registered in their estimand wording but operationalized post hoc, with every deviation logged in Appendix E. The result

¹The reported conditional CI half-width gives a retrospective, post-hoc normal-approximation resolution of ≈ 4.57 pp; it is not a shipped or preregistered MDA80 and is distinct from the canonical/strict secondary rulers. A same-bed regex-parser lineage sensitivity appears in Appendix D.

arm	bed	canonical	strict	formatted	rescue	rescue share	Wilson 95% CI	role
B1-32	1	604/1319	69/1319	131/1319	535	88.58%	[85.79, 90.87]	headline/J1
B2-96	2	303/356	299/356	350/356	4	1.32%	[0.51, 3.34]	J1 primary
B2-128	2	636/712	623/712	698/712	13	2.04%	[1.20, 3.47]	J3 baseline
B2-160	2	318/356	310/356	347/356	8	2.52%	[1.28, 4.88]	J3 contrast

Table 1: Headline fallback-rescue contrast on free-form, generative GSM8K items (there is no multiple-choice option set or chance-rate baseline). The explicit bed column is the first occurrence of each arm name and fixes the cross-bed reading: B2-96/B2-128/B2-160 are Bed 2 dose arms, whereas the numeral after B elsewhere indexes a configuration-cell family rather than a bed (Table 2). Canonical is fallback-inclusive; strict removes fallback; rescue share is rescue/canonical. The Bed 1 row compared with the three primary Bed 2 rows yields $88.58/2.52 = 35.2\times$ to $88.58/1.32 = 67.1\times$. Wilson endpoints use one half-up rounding rule; the two additional Bed 2 dose arms are included in the complete J2 inventory (Table 14), not silently generalized into this three-arm range.

is deliberately asymmetric: level remains inadmissible as a capability ordering; order must be read under both the raw-sign and favorable paired-gate rules; and the registered tie pattern is not identified from the shipped summaries on its primary estimand—the data cannot decide, which is not a proof of no effect.

2 Beds, arms, and rulers

An evaluation bed is a frozen generator, dataset, decoding configuration, and judge. Our two beds share one generator: LLaDA-8B-Instruct, a discrete masked diffusion language model (Nie et al., 2025) (official checkpoints GSAI-ML/LLaDA-8B-Instruct). The generation records are frozen artifacts from the cited trials; decoding budgets are listed per arm below, and sampling seeds and temperatures are not recorded in the frozen artifacts, so we disclose that as a provenance gap rather than retrofit values our records cannot certify (§6). Two-bed independence here is split-disjointness plus a different manipulated axis, not a cross-model claim; a cross-generator law is out of scope.

Bed 1. The LLaDA generator evaluated the 1,319-item generative GSM8K test split (Cobbe et al., 2021). Although earlier project records used the alias “GSM8K-MC,” the frozen source contains the original free-form questions and no option set; no multiple-choice wrapper, option count, or $1/K$ chance baseline is therefore defined. We retract the earlier “above-chance” wording and assess the two extraction channels only against each other. Its arms differ in their generation token budget: the matched arms B1-32/B1-64/B1-128/B1-256 use exactly 32, 64, 128, and 256 generation tokens (cells b1_matched_*), and B0-256 is a fixed-256 control (b0_fixed_256). Three holdout arms probe budgeted alternatives: H-PRED-95 decodes to a prediction-side final length of 95 tokens; H-DAEDAL-1024² decodes with the DAEDAL (Li et al., 2025) training-free dynamic length-expansion rule under a 1024-token ceiling; and H-VAR stops at a variable per-item budget. Artifact-key mappings are confined to the Appendix A provenance ledger.

Bed 2 (train-short-B). The same LLaDA generator on the GSM8K training split short-band ($n = 712$ items; length coefficient of variation 0.049), evaluated autoregressively. Its arms are dose arms spanning five token budgets—64, 96, 128, 160, and 192—from two paired audit trials: one supplies the 96/160 arms and reuses the fixed-128 run, while the other supplies the 64/192 arms (each half-arm has $n=356$). Exact trial and artifact identifiers are confined to Appendix A. The level and pattern analyses center on B2-96/B2-128/ B2-160; all five dose arms enter the J2 inventory. A whitelist-robustness property matters for the fallback channel on this bed: every unformatted Bed 2 row is also strict-wrong, so stripping the fallback changes nothing on the auxiliary strict numerator; the constructional part of

²Historical project records call this arm B2-DAEDAL (Bed 1); the H- prefix prevents collision with Bed 2 dose-arm names.

symbol	definition	symbol	definition
F, C	explicit-format and strict-correct events	q, a, s	$P(F)$, $P(C F)$, and $P(C \cap F) = qa$
m, B	discordant-pair count and bootstrap replicates	u	unformatted B2-128 items on J3 common support
F_{strip}	definite canonical relations whose strict sign reverses	MDA80	minimum detectable absolute effect at 80% power
$\Delta_{\text{can}}, \Delta_{\text{strict}}$	arm-A minus arm-B gaps under canonical/strict rulers	Δ_{true}	J3 primary conditional gap, $a_{160} - a_{128}$
b	$P(\text{canonical correct} \neg F)$, the fallback channel	S1, S2	tight-correct / raw-whitelist strict-cell sources

Table 2: Compact vocabulary used throughout. All Δ values are raw percentage points; “true” distinguishes the primary conditional estimand from the fallback-inclusive canonical ruler, not a causal effect. The numeral after B indexes a configuration-cell family, not a bed: Bed 2 dose arms use B2-⟨size⟩, whereas the historical B2-DAEDAL name denotes a Bed 1 holdout (renamed H-DAEDAL-1024 here).

this invariance is described in §6, and its data-dependent remainder must be retested on a new bed.

Rulers. The canonical judge v1.1 checks candidates in priority order: the scaffold `###`, then `\boxed{}`, then a final-position fallback that rescues the last numeric span. The strict ruler accepts only the first two (the hash/boxed whitelist). A rescue is an output judged correct by the canonical ruler only via the final-position fallback. Single-proportion intervals throughout this paper are Wilson intervals; two-proportion differences use normal intervals.

2.1 Related work and closest-work map

The beds use GSM8K-derived items (Cobbe et al., 2021) and LLaDA (Nie et al., 2025); dLLM generation and acceleration methods change decoding or compute rather than the downstream answer parser (Zhou et al., 2026; Ye et al., 2025; Li et al., 2025; Wu & Kong, 2026; Wu et al., 2025b;a). Zheng et al. (Zheng et al., 2023) belongs to the separate LLM-as-judge thread and is not evidence for deterministic extraction.

The closest extraction work is Jo et al. (Jo et al., 2025), who sweep judges and prompts over reasoning traces and show that where an answer is found can change reported performance. xFinder (Yu et al., 2024) directly targets key-answer extraction and reports failures of regex-based evaluation harnesses. Our narrower delta is empirical: we isolate a frozen parser-fallback rescue path, decompose its reported score, and compare the xFinder regex tier on the same two beds; we do not test xFinder’s learned tier or Answer Regeneration. In selective prediction, coverage and conditional accuracy are standard paired quantities in the risk-coverage trade-off (El-Yaniv & Wiener, 2010). Equation 3 is therefore a probability identity, not a novelty claim; the contribution is its registered co-report and its use to audit this specific parser artifact. Table 3 is the positioning boundary: it anchors each component to verified Jo/xFinder/selective-prediction precedents and does not turn the regex-tier reparse into a same-bed comparison with xFinder’s learned tier or Answer Regeneration.

The inferential pieces are likewise standard. (i) Format-emission base rate. The format share $q = P(\text{fmt})$ and Wilson score intervals (Wilson, 1927) summarize how often the judge must fall back; exact binomial limits are from Clopper and Pearson (Clopper & Pearson, 1934). (ii) Correctness given format. Conditional contrasts $a = P(\text{correct} | \text{fmt})$ are judged per-pair with McNemar’s paired test (McNemar, 1947), with item-paired bootstrap intervals; we explicitly decline a post-hoc equivalence claim, so the post-hoc 0.5 pp tolerance is reported without a formal TOST (Schuirmann, 1987), and family-wise control across the tested definite pairs uses Dunn’s multiple-comparison correction (Dunn, 1961). Conditioning on format emission—a post-generation property of the scored string—opens a collider-type, selected-population contrast (Elwert & Winship, 2014), which is why J1 is scoped as descriptive rather than a capability claim. (iii) Evaluation hygiene. Answer-extraction rules

Component	Closest prior work	Difference here	Class	Evidence
Extraction changes reported scores	Jo et al.; xFinder (Jo et al., 2025; Yu et al., 2024)	Frozen fallback-rescue path plus same-bed regex-lineage audit	standard phenomenon; artifact-specific audit	Tables 1, 17
q , a , and $s=qa$	Selective prediction (El-Yaniv & Wiener, 2010)	Registered co-report before any capability reading; identity itself is standard	standard identity; executed audit	Table 13
Paired order audit	McNemar and multiplicity control (McNemar, 1947; Dunn, 1961)	Raw-first rule-pair; gate pre-registered in estimand wording, operationalized post hoc	standard tests; reporting protocol	Table 6
Deviation ledger	Forde & Paganini (Forde & Paganini, 2019)	End-to-end executed instance with falsification criteria	standard practice; artifact audit	App. E
Selected-population level block	Collider literature (Elwert & Winship, 2014)	Measured two-bed rescue-share gap and its consequence	standard theory; artifact audit	Table 1

Table 3: Closest-work comparison at component level. “Standard” denotes a component already present in cited work; “artifact audit” denotes a property of these frozen beds. No priority is claimed beyond this verified neighborhood.

measurably move reported math-reasoning scores (Jo et al., 2025; Yu et al., 2024), and our preregister-then-report workflow follows calls for ML preregistration (Forde & Paganini, 2019). Within this neighborhood our narrower contribution is the auditable factorization $s = qa$ and the reported rule-pair for one fallback parser; we make no priority claim.

Claim-to-closest-work map. The level claim instantiates standard selection reasoning (Elwert & Winship, 2014); only its measured magnitude is artifact-specific. The order tests are standard (McNemar, 1947; Dunn, 1961); the raw-first rule-pair is our reporting choice. The pattern claim instantiates the extraction-sensitivity phenomenon (Jo et al., 2025; Yu et al., 2024); its not_identified verdict is specific to Table 8. Finally, the audit discipline follows Forde & Paganini (2019), while the executed deviation ledger is an artifact of this study.

Figure 1 summarizes the measurement flow and keeps the three audit conclusions visually separate; it is conceptual rather than numerical.

3 Judge normalization

3.1 Events, estimands, and the exact identity

The observational unit is an item-level binary record. Let F denote that the answer uses an accepted explicit format, here the whitelist judge_format $\in \{\text{hash}, \text{boxed}\}$, and let C denote a correct strict judgment. We report two estimands:

$$a \equiv P(\text{correct} \mid \text{fmt}), \quad \text{primary, conditional on the scored format channel,} \quad (1)$$

$$s \equiv P(C \cap F), \quad \text{auxiliary, fallback-stripped over all items.} \quad (2)$$

Writing $q \equiv P(\text{fmt})$ gives the exact factorization

$$s = qa. \quad (3)$$

For reconstruction of the fallback-inclusive ruler, define $b \equiv P(\text{canonical correct} \mid \neg F)$; then canonical accuracy is $qa + (1 - q)b$. This descriptive identity does not make b part of the strict joint estimand s . The two estimands are complementary, not replicas. The conditional a describes correctness within the formatted subset, whereas s charges a system for failing to emit the format. s also charges a system for every answer the fallback rescued: on B1-32

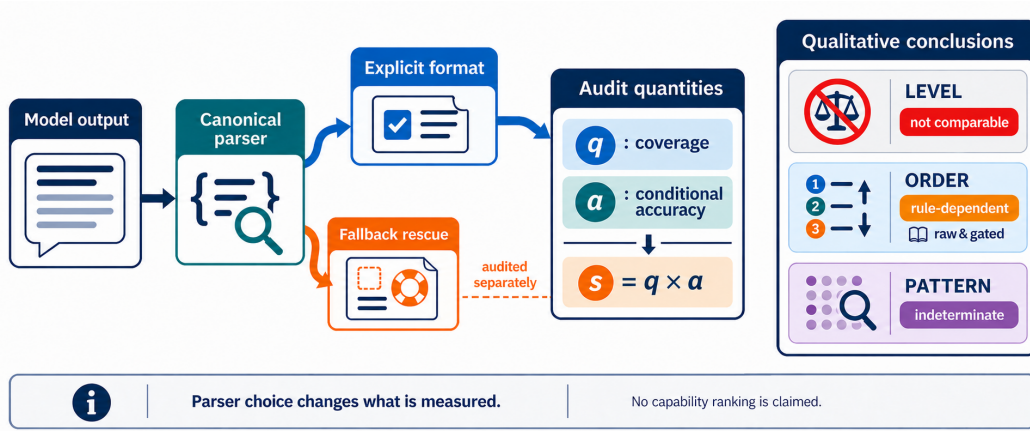


Figure 1: Conceptual overview; not quantitative evidence. Both beds enter the same canonical judge. The format audit exposes explicit-format and fallback-rescue channels before separating the three audit conclusions: level is not comparable; order leads with the raw rule and co-reports a favorable paired gate pre-registered in estimand wording but operationalized post hoc; and pattern is not identified on the registered primary estimand. The badge reading “indeterminate” denotes exactly that verdict: the data cannot decide from the shipped summaries, which is not a proof of no effect. No counts are encoded; all measurements appear in Figure 2 and Table 9.

it counts all 535 fallback-corrected answers as wrong. Across Bed 1 the fallback channel is only 5.8–9.5 pp below the format channel (§5.2); on a fallback-dominated arm, reporting both factors of $s = qa$ is therefore needed to expose whether a canonical level is driven by answer correctness, format emission, or rescue.

3.2 Why conditioning does not create a common scale

Conditioning removes the fallback channel but changes the population. Format emission is a post-outcome (post-generation) behavior: the judge decides F and C from the same generated string. Selecting on F can therefore induce a selected-population (collider-type) contrast when response properties associated with correctness also affect explicit formatting; it is not clean pre-treatment stratification. Across Bed 1 arms, $q \in [0.0993, 0.2661]$; on Bed 2, $q \in [0.9747, 0.9831]$. Thus Bed 1’s conditional accuracy describes a formatted subset comprising roughly 10–27% of its items, while Bed 2’s describes nearly all items. Unless formatting is ignorable or common support is restored, comparing the two a values is a selected-population contrast. It is useful as a sensitivity diagnostic but not as a pure capability scale. The beds also differ in prompt and task, adding an inseparable task confound. Accordingly, all contrasts below are decoding-budget $\times$ task-format comparisons on a single fixed model, not comparisons of model capability.

3.3 Predicate correction and provenance

An earlier implementation used the truth value of `judge_format`. Because the sentinel string `none` is nonempty, this predicate failed open and made $q = 1$. The frozen analysis uses whitelist membership instead. The formatted and strict columns of Table 1 give its independent recovery for all three Bed 2 arms. The auxiliary numerator is unchanged here because every unformatted row is also strict-wrong; that property is at least partly constructional (the strict numerator is a whitelist-conditioned count, §6), and the data-dependent remainder is not a general property of fallback stripping.

4 Analysis plan and disclosed revisions

Table 4 inventories the frozen supports used by the primary comparisons. Critically, Bed 2’s contrast arms are deterministic slices of one pool rather than independent runs: B2-128 is the full pool, while B2-96 and B2-160 are its under- and over-halves. J3 is recomputed on the common over-half (§5.4).

Fixed and revised rules. J1’s conditional estimand and the J1/J3 primary-arm roles were fixed before inspection; J3’s 0.5 pp tie tolerance was adopted post hoc and carries no preregistered authority. J1’s numerical residual remains descriptive because the selected populations differ. J3 uses the primary conditional estimand and is recorded as not identified from the shipped summaries when its point gap is outside the tolerance but its paired interval contains zero at much coarser resolution; failing to reject that null is not proof of no effect, and the reading is not a post-hoc TOST. The preregistration specified paired McNemar for J3 and an $m \geq 25$ disclosure heuristic inherited from the project’s earlier discordant-pair discipline; 25 is neither a theorem nor a calibrated error guarantee. The common-support paired analysis satisfies it. J1/J1a/J3 are not multiplicity-corrected and no joint error rate is claimed.

Test	Arm	Scored support	Formatted	Role / selection and resolution
J1	B1-32	1,319	131	Full frozen Bed 1 endpoint.
J1	B2-96	356	350	Under-half endpoint;
				independent-design MDA80 13.32 pp.
J3	B2-128	712; 356 common	351 common	Baseline aligned to over-half.
J3	B2-160	356	347	Paired contrast; secondary MDA80
				4.72 pp (canonical) / 4.98 pp (strict);
				primary conditional MDA80 not
				identified, with a retrospective
				CI-derived resolution ≈ 4.57 pp.

Table 4: Frozen supports for the primary J1 and J3 comparisons. Bed 2 has $CV = 0.049$; B2-128’s under- and over-halves score $322/356 = 0.9045$ and $314/356 = 0.8820$, a 2.25 pp split that makes support alignment necessary. MDA80 uses $(z_{.975} + z_{.80})\widehat{SE} = 2.802\widehat{SE}$ for a two-sided $\alpha = .05$, 80%-power normal approximation: J1 uses the observed independent-binomial variance; J3’s two reported resolutions use observed canonical/strict paired discordance. The shipped summaries do not identify a registered MDA80 for the primary conditional ratio estimand. A retrospective, post-hoc normal approximation from its reported CI half-width is $2.802(3.20/1.96) \approx 4.57$ pp; this CI-derived resolution is not a shipped or prospective design value. These are observed-variance resolutions, not prospective design guarantees.

J2 differs: its estimand wording (“definite”) was pre-registered, but the paired significance criterion was operationalized post hoc after data inspection, in the favorable direction. The raw-sign rule is therefore the lead reading and the paired gate a favorable rule-sensitivity analysis. Pairing is statistically appropriate for shared items; its post-hoc status, not its paired construction, is the disclosure. All displayed McNemar values are raw, two-sided p values; the Dunn–Bonferroni family-wise convention is a separate multiplicity sensitivity and is not mixed into the post-hoc gate’s $p < .05$ eligibility rule. The final synthesis was likewise pre-registered in estimand wording but operationalized post hoc as a level/order/pattern hierarchy. Both revisions are itemized in Appendix E. The same-bed regex-lineage sensitivity is reported separately in Appendix D; it does not redefine these registered rulers.

All effects are raw percentage points. Table 11 maps every headline value to its frozen artifact and exact machine field; all hashes and internal identifiers are confined to Appendix A.

5 Results: what survives and what vanishes

Figure 2 collects the data-derived normalization contrasts used by the result sections below.

5.1 J1: the level residual is real but not comparable

B1-32 has $a = 0.5267$, whereas B2-96 has $a = 0.8543$ (per-arm values in Table 13). Their difference is -32.76 pp (SE 4.75 pp, 95% CI $[-42.07, -23.45]$), plotted on its own scale in Figure 2(b) and audited in Table 9, so the interval excludes zero and the registered 5 pp neighborhood. Normalization therefore does not erase the numerical residual. It also does not turn that residual into a capability ordering: the comparison conditions on $q = 0.0993$ versus $q = 0.9831$, hence on sharply different item populations, and task difficulty is confounded.

The auxiliary joint quantity reinforces the channel diagnosis. Table 13 makes the anchor-free log identity primary and reports additive attribution as an anchor-dependent range. Neither is causal: the correctness component still contrasts non-common selected populations.

We therefore do not report a scale discrepancy: the additive share is not identified, its identified range $[59.11\%, 95.87\%]$ contains the unique anchor-free log share (82.58%), and format-dominance survives on every identified line. The log share is anchor-free only when the target is exactly multiplicative and factors are positive; pp-level additive shares are reported as a range, the log share as the anchor-free complement—neither substitutes for the other. Pre-registered falsification criterion: if, after this revision, the technical-soundness and experiments-and-statistics seats of the next panel still name the Table 13 decomposition rule as defective, this entire decomposition block demotes to an appendix sensitivity analysis.

5.2 The format channel is 5.8–9.5 pp higher on every Bed 1 arm—far too little to explain a 35–67× rescue ratio

The experiment’s headline states a share of canonically correct answers, not the fallback channel’s quality. Table 5 separates those quantities from the same frozen records: on B1-32, the fallback and formatted channels have accuracies 0.450 and 0.527, respectively. This supports only a channel-precision reading on the shipped items: the strict-accuracy columns are the parser-precision proxy in Table 5, reconciled by source in Table 12; parsing is a deterministic regex/Judge-v1.1 priority match with no free-text LLM scoring. The statement is bounded to Bed 1 free-form generative GSM8K, which has no multiple-choice option set, so no option-count or chance-rate number is claimed. The fallback path is therefore productive under this reference-match proxy, not established as human-labelled precision/recall. The table gives this comparison on all eight reproducible Bed 1 arms, using the identity $\text{strict} + \text{rescue} = \text{canonical}$ (verified arm-by-arm from the per-arm accuracy table; every rescued row is unformatted and every format-correct row is strict-correct, a property we revisit in §6).

Both channels classify many frozen items correctly, but the format channel is consistently 5.8–9.5 pp higher (5/8 uncorrected-significant under S1 and 6/8 under S2; 3–4/8 Bonferroni across the two frozen strict-cell sources; Table 5). That modest directional gap is real evidence against channel equivalence, yet it is far too small to explain the 35–67× rescue-share contrast. The contrast is therefore prevalence-dominated: q is low on Bed 1 and near one on Bed 2. The gate makes that distinction visible by requiring q , a , and s together; the strict ruler’s treatment of all 535 rescued B1-32 answers as wrong is not a clean capability correction.

5.3 J2: raw order first, with a disclosed paired-gate rule sensitivity

The estimand wording “definite” was pre-registered, but its criterion was operationalized post hoc after inspecting the data, in the claim-preserving direction: a canonical relation is definite only when exact two-sided McNemar has $p < .05$ and an item-paired bootstrap interval excludes zero. This is a directionally favorable rule sensitivity: the gate was pre-registered in estimand wording and operationalized post hoc. Pairing itself is appropriate because every Bed 1 arm scores the same 1,319 items; the disclosure concerns when and why the gate was chosen. The full matrix uses $B = 2000$ paired resamples. J3 uses $B = 5000$ for its single primary interval; each count was fixed in its analysis script before resampling, and the unequal computational allocation carries no evidentiary ranking.

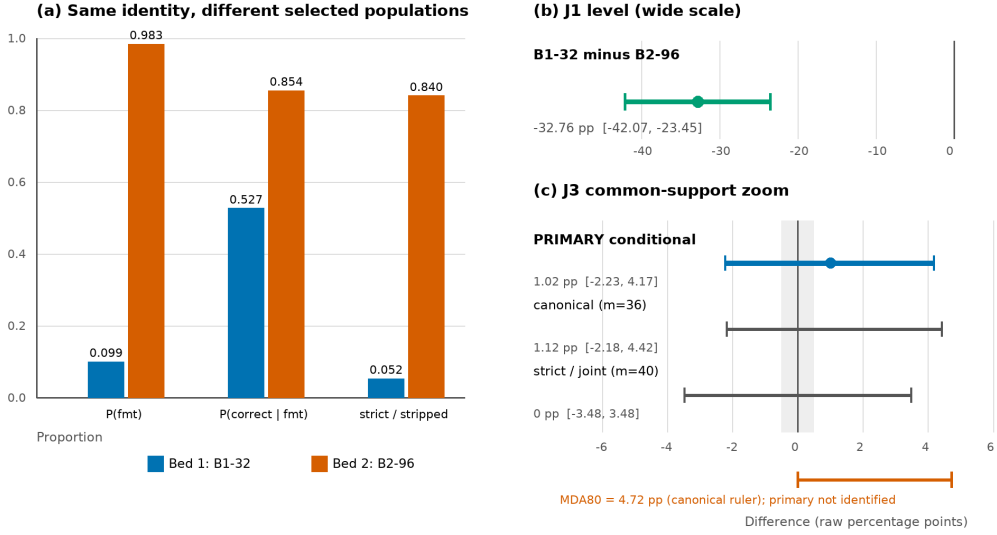


Figure 2: Machine-derived normalization summary. (a) The same identity holds on both beds despite sharply different format coverage. (b) J1 is shown on its own wide scale because it lacks a common selected population. (c) The J3 zoom makes the primary conditional estimand visually dominant; strict and canonical are secondary rulers. The shaded ± 0.5 pp explicitly post-hoc tolerance and the 4.72 pp canonical-ruler MDA80 bracket show why a point outside tolerance with an interval straddling zero is not identified as equivalent: the data cannot decide. MDA80 for the primary conditional ruler is not identified from the shipped summaries; a retrospective, post-hoc normal approximation from its reported CI gives ≈ 4.57 pp, distinct from the secondary-ruler bracket. Values come from the frozen per-item Bed 2 rerun and conditional paired analysis; the rendering script asserts their hashes.

The exact McNemar p values and the $p < .05$ eligibility rule are raw and two-sided. Dunn-Bonferroni family-wise control is reported as a separate multiplicity sensitivity; it does not retrospectively redefine eligibility.

Accordingly J2 is reported as a raw-first rule-pair, not a unified pre-registered verdict: the raw point-estimate rule reads $3/28$, while the paired significance gate reads $F_{\text{strip}} = 0/23$ as a favorable, post-hoc rule sensitivity; each looser definite gate has two reversals. Table 6 keeps the non-interchangeable raw, paired-definite, and planned-arm denominators together; the complete Bed 1 matrix is in Appendix C.

The shipped Bed 1 records identify only eight arms (28 pairs), although the frozen plan asserts nine arms (36 pairs); the ninth arm is unidentifiable from the shipped records. Under an adversarial assignment of plausible reversals, the bound is $8/31$ on the gated support and $11/36$ on the raw 36-pair plan; both denominators are carried in the last row of Table 6.

The gate is near-collinear with the reversal mechanism it conditions on: definite-by-construction excludes the ambiguous pairs. Table 7 makes that conditioning visible and co-reports the frozen arm-resampling and sharp sign-null diagnostics. Every observed raw reversal lies in the smallest-gap stratum removed by the paired gate; the gated zero is therefore a conditional sensitivity reading, not evidence that raw order is universally preserved.

The complete Bed 2 denominator is derived item by item in Table 14: four fixed-128 baseline contrasts are pairable, two are non-definite, and therefore eligible $= 4 - 0 - 2 = 2$.

Thus the evidence is deliberately dual: three reversals among all 28 raw signs, but none among 23 definite Bed 1 pairs under the paired gate that was pre-registered in estimand wording, operationalized post hoc, and favorable in direction; Bed 2's two-pair definite inventory is too small to support a broad zero-reversal law.

arm	n_{fmt}	$a=P(\text{corr} \mid \text{fmt})$	$P(\text{corr} \mid \neg\text{fmt})$	Δ (pp)	z (S1/S2)
B1-32 (J1)	131	0.526718	0.450337	+7.64	1.662 / 1.662
B1-64	149	0.664430	0.605983	+5.84	1.417 / 1.417
B1-128	160	0.806250	0.723037	+8.32	2.455 / 3.445
B1-256	348	0.824713	0.745623	+7.91	3.200 / 4.254
B0-256	351	0.831909	0.742769	+8.91	3.652 / 4.362
H-PRED-95	147	0.775510	0.709898	+6.56	1.779 / 2.006
H-DAEDAL-1024	165	0.848485	0.758232	+9.03	2.947 / 2.947
H-VAR	137	0.540146	0.445008	+9.51	2.116 / 2.116

Table 5: Per-channel classification accuracy on the eight reproducible Bed 1 arms (source: the per-arm accuracy table; $n=1319$ items/arm; z uses the unpaired two-proportion formal interval). The format channel (fmt) exceeds the fallback channel ($\neg\text{fmt}$) on all eight arms under S1 (5.8–9.5pp; the direction also holds on all eight under S2): five of the eight S1 differences clear $z=1.96$ at $\alpha=0.05$ (B1-128, B1-256, B0-256, H-DAEDAL-1024, H-VAR), and six do under S2, which adds H-PRED-95. The final column reports every arm under S1 (tight-correct, used by all paper computations) and S2 (raw-whitelist alternative), in that order. At the Bonferroni boundary $z=2.7339$, the tally is therefore a source range of 3–4/8: 3 under S1 and 4 under S2. B1-128 crosses only under S2; its S1 $z=2.455$ lies between the uncorrected 1.96 and Bonferroni 2.7339 boundaries and needs +2 flips (+1.25 pp on a_{fmt}) to cross. The dual values are reproducible from results/r428_dual_source_sweep.json (validation against the frozen c3_tight accuracies at 10^{-5}). Four cells have a nonzero S1-versus-S2 formatted-correct count—B1-128: 129 vs. 133; B1-256: 287 vs. 293; B0-256: 292 vs. 296; H-PRED-95: 114 vs. 115 (the first two are the reviewer-queried pair)—and each is an item-level extractor-fidelity difference where the canonical extractor rescued a correct answer the strict extractor refused; the other four arms agree exactly, so their two z values coincide. Only B1-128 changes Bonferroni status between sources. For the remaining sub-threshold S1 cells, H-VAR needs 4 flips; H-PRED-95 needs 5; B1-32 needs 6; B1-64 needs 7. No other cell is within two flips of the boundary. For B1-32, the displayed channel accuracies are $69/131 = 0.527$ (formatted) and $535/1188 = 0.450$ (unformatted), while Table 1 reports the distinct $535/604 = 88.6\%$ rescue share. The 8-for-8 direction tally is descriptive: the arms share the same 1,319-item pool and generator, so it is not a claim of eight independent Bernoulli replicates. An item-clustered permutation would be the inferential form and is not claimed here; bootstrap intervals and paired tests elsewhere cluster by problem ID. On the J1 arm alone the difference is not significant ($z=1.66$). The 88.6% headline ratio is driven by the low format-emission base rate ($q=0.0993$ on B1-32), not by rescue being a noise source.

5.4 J3: the registered tie, recomputed on common support

The full-pool canonical tie is an artifact of comparing supports of different sizes. We therefore align B2-128 and B2-160 on the common over-half and report the strict, canonical, and conditional rulers separately in Table 8.

The strict row is a genuine exact tie but remains auxiliary: the preregistration assigns J3 to the primary conditional ruler. The canonical row is fallback-inclusive over all common-support items, not conditional on format. Direct enumeration of the frozen per-item Bed 2 rerun resolves the unformatted B2-128 count as $u=5$ (previously $u \in \{4, 5\}$), producing the conditional row rather than collapsing distinct rulers into one number. The exact fields and artifact hash appear in Appendix A; the same records support the conditional paired bootstrap.

The registered primary point estimate crosses the explicitly post-hoc center tolerance, but its paired interval contains zero and the detectable resolution reported in the caption is much coarser. J3 is therefore not identified from the shipped summaries: the data cannot decide, and failing to reject the null is not proof of no effect. We neither claim a broken tie nor retain the exact-tie label. The old full-sample result conflated the rulers and populations and is retracted in §5.5. As always, an interval containing zero is not evidence of equivalence; resolving the post-hoc tolerance would require a much larger paired sample.

Reading	Eligible	Reversals	Item-paired strata	Role / observed sensitivity
Bed 1 raw sign	28	3	3/4, 0/4, 0/20	Lead point-estimate rule; eight of nine arms identifiable.
Bed 1 paired canonical-definite	23	0	0/3, 0/20	Favorable rule sensitivity; estimand wording pre-registered, criterion operationalized post hoc.
Merged gate set	25	0	—	Bed 1 gate plus two definite Bed 2 baseline pairs.
Bed 2 paired canonical-definite	2	0	—	Power-limited inventory, derived in Table 14.
Bed 1 strict-definite	22	2	—	Looser definite gate; two reversals.
Bed 1 either-definite union	25	2	—	Looser definite gate; the same two reversals.
Adversarial planned-arm bound	31 gated / 36 raw	at most 8 / 11	—	Missing ninth arm assigned the reviewer-most-adverse plausible reversals.

Table 6: J2 rule-pair and looser gates. The two looser-gate reversals are B1-256-H-DAEDAL-1024 and B0-256-H-DAEDAL-1024, both canonical near-ties whose paired intervals include zero; the raw rule adds a third near-tie. Every zero here is a descriptive count, not a probability: this revision withdraws the independent-Bernoulli zero-event upper bounds this column previously carried, because all pairs are built from the same shared problem IDs and are therefore not independent trials (the withdrawn values are itemized in Appendix E). The replacement column reports the item-paired $|\Delta_{\text{can}}|$ strata of Table 7 in the order <1 pp, $1 \leq |\Delta_{\text{can}}| < 4$ pp, ≥ 4 pp, which localizes where the reversals actually sit rather than converting a zero into confidence. Exact McNemar p values and gate eligibility are raw and two-sided; Dunn–Bonferroni family-wise control is a separate sensitivity. The favorable paired gate—pre-registered in estimand wording and operationalized post hoc—is near-collinear with the reversal mechanism: its criterion conditions away the ambiguous near-ties where every observed raw reversal occurs.

Diagnostic	Support	Result	Takeaway
Raw gap $ \Delta_{\text{can}} < 1$ pp	4 pairs	3 reversals	All observed raw reversals are near-ties.
Raw gap $1 \leq \Delta_{\text{can}} < 4$ pp	4 pairs	0 reversals	No reversal in the middle stratum.
Raw gap $ \Delta_{\text{can}} \geq 4$ pp	20 pairs	0 reversals	No reversal in the widest-gap stratum.
Strict-channel arm permutation	2,000 draws	0 draws with $F_{\text{strip}} \geq 1$	Canonical pairing fixed; definite count clusters at 21–23.
Canonical-channel arm permutation	2,000 draws	0 draws with $F_{\text{strip}} \geq 1$	Strict pairing fixed; definite count clusters at 21–23.
Sharp sign exchangeability	23 definite pairs	$P(F_{\text{strip}} = 0 \mid H_0) = 2^{-23} \approx 1.2 \times 10^{-7}$	Fair-coin diagnostic for each definite pair’s strict sign.

Table 7: Gate-selection diagnostics for J2. Gap rows stratify the frozen 28-pair raw matrix; null rows summarize the two single-channel arm-permutation checks and the sharp sign-exchangeability calculation. The permutation script reproduces every arm’s frozen canonical/strict accuracy to 10^{-5} ; artifact paths are in Appendix A.1. These are null diagnostics, not population reversal guarantees. Within the favorable paired-definite subset, no below-1 pp pair is eligible; the two wider strata contain 0/3 and 0/20 reversals. All counts share problem IDs and are descriptive rather than independent-Bernoulli probability statements.

Ruler	B2-128	B2-160	Δ (pp)	Paired evidence and role
Strict / joint <i>s</i>	310/356 = 0.8708	310/356 = 0.8708	0.00	$m=40, p=1$, CI $[-3.48, 3.48]$; exact tie, auxiliary.
Canonical	314/356 = 0.8820	318/356 = 0.8933	+1.12	$m=36, p=0.6177$, CI $[-2.18, 4.42]$; fallback-inclusive sensitivity.
Conditional $P(\text{correct} \mid \text{fmt})$	310/351 = 0.8832	310/347 = 0.8934	+1.018	Item-paired bootstrap CI $[-2.23, +4.17]$ ($B=5000$); primary marginal conditional estimand, not identified from the shipped summaries. Equal numerators ($310 = 310$) make this gap entirely a four-item denominator effect.
Joint-format sensitivity	304/342 = 0.8889	306/342 = 0.8947	+0.585	Both-formatted support; McNemar $m=30, p=0.8555$.
Superseded conditional mixed support	—	—	+0.08	CI $[-4.06, 3.90]$; re-signed to the current $a_{160} - a_{128}$ convention; source fractions omitted because the populations differ.

Table 8: J3 estimands after support alignment. The superseded mixed-support row is retained only for provenance and re-signed from the legacy $a_{128} - a_{160}$ convention to the current $a_{160} - a_{128}$ convention; its source fractions (623/698 vs. 310/347) remain in Appendix A because the populations differ. The auxiliary strict ruler ties exactly, whereas the registered primary conditional gap is +1.018 pp with a paired interval that contains zero; its equal numerators ($310 = 310$) are a frozen-data coincidence, and the gap is a four-item denominator effect (351 vs. 347 after resolving $u=5$). Canonical/strict MDA80 values are 4.719/4.977 pp; primary conditional MDA80 is not identified from the shipped summaries. A retrospective, post-hoc normal approximation from the conditional CI half-width gives $2.802(3.20/1.96) \approx 4.57$ pp; it is a CI-derived resolution, not a shipped or preregistered MDA80. The arm-wise conditional point estimates (310/351 and 310/347) and the joint McNemar support ($n = 342$) are different subsets. The +1.018 pp arm-wise gap is entirely a denominator effect because the numerators are equal ($310 = 310$); the $n = 342$ intersection supports the separate joint-format test. At $n_{\text{selected}} = 347$, two flips drop the point below the explicitly post-hoc 0.5 pp tolerance, four reduce it to zero, and eight lift the 95% CI lower bound above zero (results/r428_j3_oneflip.json). Thus the primary result remains not identified from the shipped summaries—the data cannot decide—not an equivalence or robustness-to-zero claim. Paired bootstrap and exact-test units are problem IDs whenever records share question identifiers.

5.5 Deviations from the preregistration

Appendix E itemizes every departure from the pre-data record, including the J3 population/estimand, the order gate whose wording was pre-registered but criterion operationalized post hoc, the J1 capability reading, and the synthesis operationalized post hoc, with each directional effect.

A hierarchy of two-bed claims. The evidence supports a hierarchy, not a binary “comparable / not comparable” verdict; Tables 9, 19, and 20 carry every value quoted in this paragraph beside the result it qualifies. A level claim needs a common scale and common selected population; neither is available. The raw order rule has three reversals among 28 pairs, while the favorable paired-gate rule sensitivity—pre-registered in estimand wording and operationalized post hoc—has no definite reversal among 23 pairs. A pattern claim asks whether a registered tie changes its label; on the pre-registered primary estimand J3 is not identified from the shipped summaries—the data cannot decide: its +1.02 pp conditional gap exceeds the explicitly post-hoc 0.5 pp tolerance, but the item-paired 95% CI contains zero, so the data cannot decide—while an exact tie persists only under the auxiliary strict estimand. MDA80 for the primary conditional ruler is not identified; the canonical/strict secondary rulers have resolutions of 4.719/4.977 pp, while the primary conditional CI gives only a retrospective, post-hoc ≈ 4.57 pp resolution. The hierarchy therefore

co-reports the favorable gated zero with raw and looser-gate reversals; it does not promote that zero into a universal order law.

ID	Question	Frozen observation	Decision and scope
J1	Two-bed level after conditioning	−32.76 pp, CI [−42.07, −23.45]; $q = 0.0993$ vs. 0.9831; design-entailed	not comparable: non-common selected populations plus task confounding.
Joint	Fallback-stripped level	−78.76 pp, CI [−82.75, −74.76]; additive share [59.11%, 95.87%]; log share 82.58%	Additive point attribution is not identified; log is anchor-free and exact for positive multiplicative factors.
J2	Definite order reversal	Bed 1: 3/28 raw, 0/23 disclosed paired, and 2 under both looser gates; Bed 2: 0/2	Raw-first rule-pair; the favorable paired gate was pre-registered in estimand wording but operationalized post hoc. Descriptive raw strata are 3/4 below 1 pp, 0/4 at 1–< 4 pp, and 0/20 at ≥ 4 pp; shared problem IDs preclude an independent-Bernoulli probability reading.
J3	Registered tie after normalization (common support)	strict tie (310 = 310, $m=40$, $p=1$); canonical +1.12 pp; primary marginal conditional +1.018 pp, joint-format +0.585 pp	Primary not identified from the shipped summaries: the data cannot decide because CI [−2.23, +4.17] pp contains zero. Primary MDA80 is not identified; canonical/strict secondary values are 4.719/4.977 pp, and the primary CI-derived post-hoc resolution is ≈ 4.57 pp. Strict tie is auxiliary; no TOST.
Final	Synthesis	J1 level blocked; J2 is rule-dependent; J3 tie not identified on its primary estimand	Pre-registered in estimand wording, operationalized post hoc as a hierarchy: the raw rule has reversals while the favorable paired gate reads zero; pattern is not identified from the shipped summaries and level is inadmissible.

Table 9: Decision audit, distinguishing pre-registered elements from post-hoc revisions. Effects use raw pp; Add./Log. are defined in §5.

The gate. For any conclusion whose canonical score can use fallback: (i) report $P(\text{fmt})$, $P(\text{correct} \mid \text{fmt})$, and their joint product; (ii) state formatted-subset coverage on every bed; (iii) retest the relation after fallback removal; and (iv) reject a two-bed level law unless common support and task-scale comparability are separately justified. Passing the gate permits only the relation actually tested. It never upgrades an observed zero reversal into a universal law.

6 Limitations and validity threats

This audit spans only two beds, two rulers, and different tasks/prompts. $P(\text{correct} \mid \text{fmt})$ conditions on arm-varying post-outcome (post-generation) behavior, so level contrasts are descriptive; unregistered reweighting is unavailable. Unpaired claims use normal intervals; J3 uses the paired common support. Only two Bed 2 J2 pairs are definite, and J3 cannot resolve the post-hoc 0.5 pp tolerance. Whitelist-strict behavior is partly constructional, but formatted-row errors remain data-dependent and need retesting. No pooled two-bed SD exists; pp effects are unstandardized.

Reproducibility statement. Appendix A lists exact fields and SHA anchors, interval conventions, the numeric guard, and deterministic build. Analysis is reproducible from the shipped tables, while generation inherits the frozen-trial disclosure: seeds and temperatures absent from those records cannot be reconstructed. The canonical-item rebuild verification and its below-gate drift record are frozen alongside the artifact; Appendix A.1 gives the role-named paths and exact fidelity result.

Ethics statement. No human participants or personal data are used. The main risk—misleading capability comparison—is addressed by scoped claims, post-hoc operationalization labels, and provenance.

7 Conclusion

Fallback removal is necessary but insufficient. The format channel’s modest directional advantage is too small to explain the prevalence-dominated rescue-ratio contrast (Table 5). The order result depends on the rule: the raw-first and favorable paired-gate readings differ, and the paired gate was pre-registered only in estimand wording and operationalized post hoc (Table 6). J3 remains not identified on its primary conditional estimand, while J1 remains descriptive rather than a capability law (Tables 8 and 9). The additive attribution is likewise not identified; its legal range contains the unique log share, and no scale discrepancy is claimed (Table 13). Reporting q , a , and $s = qa$ makes these boundaries explicit.

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AI assistance disclosure. AI-assisted writing and image-generation tools were used to reorganize, typeset, and check this manuscript and to produce the conceptual Figure 1. All numerical claims and Figure 2 were restricted to the frozen machine artifact and mapped to explicit evidence IDs; no experiment or citation was created by the writing tool.

Appendix roadmap. The appendices are ordered by role rather than by order of citation, because the main text reaches the parser and deviation ledgers before it needs the provenance manifest. Appendix A is the artifact manifest and availability statement (including the per-claim provenance and strict-cell reconciliation tables); Appendix B holds the supporting numeric audits; Appendix C is the complete J2 paired definiteness inventory; Appendix D is the same-bed parser-lineage audit; and Appendix E is the full deviation list. Every appendix letter in this paper is resolved by the compiler from its label, so no cross-reference depends on a hand-written letter.

A Artifact manifest

The current quantitative evidence is results/judge_normalization/judgment_table.json, SHA-256 prefix c77ec165 (J-value set frozen as 566a1983c268); the preregistration SHA-256 prefix is 974d687224b1; full hashes are recorded in evidence/repository_inventory.md and the build audit. The analysis script, data-figure script, claim map, and method map are, respectively, scripts/judge_normalization.py, scripts/plot_judge_normalization.py, evidence/claim_evidence_map.tsv, and evidence/method_provenance.tsv. The data-figure script asserts the evidence hash before rendering.

A.1 Availability and reproducibility

An anonymized supplementary artifact bundle accompanies the submission. Its manifest pointer is `paper/.p5_iclr_manifest.json`, whose SHA-256 is `a76a7071d0ce6e6a4ec6ea72c89338eb45051e0847bfe57dcca487ace71d508`; the official-template archive recorded inside it has SHA-256 `b6d63b29992e153f804bb6d170c57db156c011b5bedf96a9f31d58813b909acf`. Analysis is fully reproducible from the shipped tables and scripts. Generation is only reproducible to the extent certified by the frozen trial artifacts: model, split, arm budgets, and outputs are present, while missing sampling seeds and temperatures are disclosed rather than reconstructed.

Availability. The anonymized supplementary bundle, the build ledger, the canonical-items rebuild verification scripts (`scripts/r419_parser_lab/parsers.py`, `scripts/r419_parser_lab/run_lab.py`), the numeric-guard script (`scripts/check_numeric_tables.py`) together with its recorded output (`results/r409_numeric_table_guard.json`), and the Figure 2 renderer are frozen with the artifact and are listed by role in the remainder of this subsection. Frozen trial artifacts are not redistributed; their role-named identifiers suffice to re-derive the numeric claims from the shipped tables alone. Consolidated role pointers are: the bundle and build ledger in this Availability paragraph; canonical parser rebuild, numeric-guard, and Figure 2 renderer paths immediately below; and the irrecoverable seed/temperature disclosure in §2 and §6.

The canonical-items rebuild verification assets are `scripts/r419_parser_lab/parsers.py` and `scripts/r419_parser_lab/run_lab.py`, with evidence in `results/r419_parser_lab/lab_results.json`. They record canonical-extractor drift $6/2136 = 0.28\%$, below the pre-registered 2% gate in `prereg/R419_PREREG_SAMEBED_NEAREST_NEIGHBOR_REPARSE.md`.

Shipped assertion suite. The cell-level numeric guard is `scripts/check_numeric_tables.py` with its recorded output `results/r409_numeric_table_guard.json`. The Figure 2 renderer asserts that the 4.719 pp and 4.977 pp MDA80 fields retain their canonical/strict ruler labels and that the primary conditional ruler is not assigned either value. The primary conditional CI-derived ≈ 4.57 pp resolution is explicitly post-hoc and is not asserted as a shipped MDA80. The denominator-convention assets are the frozen J2 rule file and paired result; together they assert 28 raw pairs, 23 gated pairs, and the separately derived ninth-arm bounds. These are descriptions of shipped checks, not claims that new experimental runs were executed in this revision.

The refine-6 J2 resampling-null assets are `scripts/r427_fstrip_resampling_null.py`, `results/r427_fstrip_arm_resampling.json`, and `results/r427_fstrip_nulls_supplement.json`; the script reconstructs every frozen arm before applying the recorded single-channel permutations.

Entry	Verified primary locator	Verification basis
Dream 7B (Ye et al., 2025)	arXiv:2508.15487	primary arXiv record, 2026-08-24
Fast-dLLM (Wu et al., 2025b)	arXiv:2505.22618	primary arXiv record, 2026-08-24
Fast-dLLM v2 (Wu et al., 2025a)	arXiv:2509.26328	primary arXiv record, 2026-08-24
DreamOn (Wu & Kong, 2026)	arXiv:2602.01326	primary arXiv record, 2026-08-24
Jo et al. (Jo et al., 2025)	arXiv:2510.14773	primary arXiv record, 2026-08-24
xFinder (Yu et al., 2024)	arXiv:2405.11874	primary arXiv record, 2026-08-24
Selective classification (El-Yaniv & Wiener, 2010)	JMLR 11:1605–1641	publisher record, 2026-08-24

Table 10: Completion of the per-entry verification ledger: the four previously unlogged dLLM references and the positioning-critical Jo et al. entry. The remaining methodology entries are covered by the bounded retrieval ledger.

All 95% confidence intervals use $z=1.96$ and one shared routine (scripts/judge_normalization.py diff_ci; Wilson intervals via plot_judge_normalization.py); endpoints are rounded half-up, and full-precision values are stored in the frozen judgment table and the per-item Bed 2 rerun rebuild_full.json paired_stats. The original generation-and-judge references used here (Cobbe et al., 2021; Nie et al., 2025; Li et al., 2025; Zhou et al., 2026; Zheng et al., 2023) were each verified against a primary arXiv record or an in-run multi-reviewer citation ledger before inclusion; Table 10 records the per-entry locator and verification basis for the entries that ledger had not yet covered. The eight methodology entries added in the bounded positioning pass are independently fetch-verified in refs_retrieval/r409_refpos_citations_verify.json. Unverified fields and failed retrievals are omitted rather than inferred from memory.

Per-claim provenance. Each quantitative claim below resolves to one artifact, its SHA-256 prefix, and the exact JSON field from which the value is read; every value is regenerated in-repo (see scripts/judge_normalization.py and scripts/plot_judge_normalization.py). Independent verification is recorded in results/r197_refine_verify/REFINE_VERIFY_R197.json (SHA-256 prefix 515a497a).

The three independent provenance artifacts are named descriptively: the per-arm accuracy table (c3_tight), the frozen strict-decomposition cells (strict_fmt_decomposition), and the per-item Bed 2 rerun (rebuild_full.json, SHA-256 prefix bfebbd1). Their exact addresses are in evidence/repository_inventory.md.

Table 11 maps each headline decision to its artifact, hash prefix, and machine field.

The frozen 566a1983c268 judgment table remains the source for the J1 level claim, but its J3_equiv_to_sig_is_artifact field (the old -0.08 pp full-sample value) is superseded for the J3 claim by the per-item rerun’s paired_stats/arm_160_{strict,canonical} common-support values and the conditional paired analysis (see §5.4 and J3 source notes), and its J2 strict-pairs inventory is superseded by R197 R2_J2_8arm_sensitivity (28 reproducible pairs, 9th arm unidentified, as disclosed in §5.5). The r402 paired artifacts govern the current J2 gate and J3 primary interval; the per-item rerun governs the common-support counts.

Table 12 reconciles the strict molecules behind the Headline row. The two cited sources are the per-arm c3_tight table and the frozen strict_fmt_decomposition cells. The authoritative dual-source manifest is results/r428_dual_source_sweep.json (SHA-256 prefix 5d6f918c): rows[/{source,n_fmt,fmt_correct,notfmt_correct,z} stores every displayed

Claim	Artifact (SHA prefix)	JSON field path	Value
Headline	REFINE_VERIFY_R197 (515a497a)	R1_headline/canon_correct; flip_rescued; rescue_rate_pct; ratio_range_x	535/604 = 88.58%; 35.2–67.1×
J1 level	judgment_table.json (frozen 566a1983, current c77ec165)	J1_crossbed_level_ after_normalize	$\Delta = -32.76$ pp
J2 rule-pair	r402 paired analysis (1fb1a156)	definite_paired_count; F_strip_definite_reversal; F_strip_raw_reversal; n_pairs	0/23 paired; 3/28 raw
J3 strict	rebuild_full.json (bfebbd1)	paired_stats/arm_160_strict	310/356 m40 p1
J3 canonical	rebuild_full.json (bfebbd1)	paired_stats/ arm_160_canonical	314/356 vs. 318/356; +1.12 pp
J3 primary conditional	r402 conditional-paired JSON (17ec0132)	conditional_estimand; paired_bootstrap/ci95_pp	310/351 vs. 310/347; +1.018 pp; $[-2.23, 4.17]$
J3 MDA rulers	r402 conditional paired + judgment table (17ec0132/c77ec165)	paired_bootstrap/ci95_pp; canonical/strict stored fields	primary MDA80 not identified; canonical/strict 4.719/4.977 pp; post-hoc CI-derived primary resolution ≈ 4.57 pp
J3 joint- format	same conditional-paired JSON (17ec0132)	both_fmt_mcnemar	304/342 vs. 306/342; +0.585 pp; $p = 0.8555$
J3 superseded	r409 Table-6 verification JSON (60d5b441)	table6_superseded_row/true_ source_recovered	stored old-order -0.082 pp; current convention +0.082 pp

Table 11: Per-claim provenance for the headline values and decision audit.

channel value, while construction/{S1_def,S2_def} fixes S1 as tight-correct and S2 as the raw-whitelist alternative.

Check	S1: per-arm c3_tight	S2: frozen raw-whitelist	Consequence
32/64-token, H-DAEDAL-1024, H-VAR cells	agree	agree	No source sensitivity; identical z .
B1-128 (n_{fmt} , correct)	(160, 129); 129/160	(160, 133); 133/160	Difference of 4 formatted-correct items.
B1-256 (n_{fmt} , correct)	(348, 287); 287/348	(348, 293); 293/348	Difference of 6 formatted-correct items.
B0-256 (n_{fmt} , correct)	(351, 292); 292/351	(351, 296); 296/351	Difference of 4 items; above the boundary under both sources.
H-PRED-95 (n_{fmt} , correct)	(147, 114); 114/147	(147, 115); 115/147	Difference of 1 item; below the boundary under both sources.
B1-256 strict accuracy over all items	287/1319 = 0.2176	293/1319 = 0.2221	Source-to-column assignment matches the item row above.
Table 5 z	B1-128: 2.455; B1-256: 3.200; B0-256: 3.652; H-PRED-95: 1.779	B1-128: 3.445; B1-256: 4.254; B0-256: 4.362; H-PRED-95: 2.006	Bonferroni count is 3/8 (S1) versus 4/8 (S2); only B1-128 switches.

Table 12: Strict-cell source reconciliation. The r420 audit identified, and this revision fixes, the prior 293/287 cross-column transposition. S1 is always the per-arm tight-correct source used by the paper’s computations; S2 is the frozen raw-whitelist alternative. The integer pairs and all-item accuracies make that assignment explicit rather than inferential. The B1-256/B0-256 crossing is a between-source point flip and not one of the 3/28 canonical-versus-strict ruler reversals: within either source B0-256 stays higher (0.2214 vs. 0.2176 under S1; 0.2244 vs. 0.2221 under S2), and the ordering flips only if a reader mixes B1-256 from S2 with B0-256 from S1. The order rule-pair is unchanged, but the channel-wise Bonferroni count is the source range 3–4/8 shown for all eight arms in Table 5.

Our earlier wording (“No number in the body depends on those molecules”) was too strong: Table 5 does depend on those cells. All J2/J3 calculations in this paper take the per-arm c3_tight values.

B Supporting numeric audits

Arm / contrast	q	a	$s = qa$	Additive-share range	Symmetric share	Log share
B1-32	0.09931	0.52672	0.05232	—	—	—
B2-96	0.98315	0.85428	0.83991	—	—	—
B1 minus B2	−0.88384	−0.32756	−0.78760	[59.11%, 95.87%]	77.49%	82.58%

Table 13: J1 per-arm co-report and decomposition. The first three columns are the frozen format coverage, conditional accuracy, and exact product. Every additive split of Δs requires an anchor: the two one-anchor rules span [59.11%, 95.87%], so the additive share is not identified. The symmetric Oaxaca choice uses $\bar{a} = 0.69050$, $\bar{q} = 0.54123$, and $\Delta s = \bar{a}\Delta q + \bar{q}\Delta a$; it is one sensitivity anchor. The primary anchor-free exact decomposition is $\Delta \ln s = \Delta \ln q + \Delta \ln a$, with log share 82.58% inside the additive range. The formerly registered residual bookkeeping (58.4%; correctness term equal to unweighted Δa), the symmetric split (77.49%), and the exact log row (82.58%) use distinct dependence estimators; this is effect-of-cause bookkeeping, not a data difference. In particular, the additive “format term” is the residual $\Delta s - \Delta a$, not a q -effect; the symmetric split gives −61.03/−17.73 pp (77.5%); and the log row is exact. The old additive row is not an identified decomposition and lies below the legal range.

candidate	item-paired baseline	pairable	m	p_{McN}	paired 95% CI (pp)	eligible / exclusion
B2-64 (Bed 2)–B2- 128	under-half, 356	yes	82	1.73×10^{-15}	$[-23.68, -14.53]$	yes; canonical definite
B2-96–B2- 128	under-half, 356	yes	41	0.0043	$[-8.82, -1.86]$	yes; canonical definite
B2-160–B2- 128	over-half, 356	yes	36	0.6177	$[-2.18, 4.42]$	no; non-definite
B2-192 (Bed 2)–B2- 128	over-half, 356	yes	33	0.0801	$[-0.06, 6.24]$	no; non-definite

Table 14: Derivation of the Bed 2 J2 denominator from the per-item rerun. The candidate set contains only fixed-128 baseline contrasts supplied as item-paired within each dose trial/reused baseline alignment; cross-dose pairs are outside this fixed-baseline inventory. The 712-item baseline is aligned to the corresponding 356-item under/over half. Thus eligible = 4 candidates – 0 non-pairable – 2 non-definite = 2. B2-192 crosses $p = .05$ only under the strict ruler (.0801 $\rightarrow$.0470) without reversing sign.

C J2 paired definiteness table

Tables 15 and 16 split the full inventory to retain readable type. All 28 Bed 1 arm pairs share the common 1,319 items. The raw-sign rule is reported first. definite uses the favorable paired gate, pre-registered in estimand wording and operationalized post hoc (exact two-sided raw McNemar $p < 0.05$ AND paired-bootstrap 95% CI excluding 0, $B=2000$); rev marks a strict-vs-canonical sign reversal (- none, = exact canonical tie, $\Delta=0$). The gate selects 23/28 pairs and reads $F_{\text{strip}} = 0/23$; the raw rule reads 3/28 (B1-32–H-VAR, B1-256–H-DAEDAL-1024, and B0-256–H-DAEDAL-1024). All three are canonical near-ties whose paired intervals contain zero.

pair	Δ_{can} (pp)	Δ_{strict} (pp)	m	p_{McN}	paired CI% (95%)	definite	rev
B1-32–B1-64	-15.47	-2.27	376	0.000	$[-18.35, -12.66]$	Y	-
B1-32–B1-128	-27.52	-4.55	525	0.000	$[-30.71, -24.49]$	Y	-
B1-32–B1-256	-30.86	-16.53	563	0.000	$[-33.89, -27.67]$	Y	-
B1-32–B0-256	-30.86	-16.91	557	0.000	$[-33.81, -27.67]$	Y	-
B1-32–H-PRED-95	-25.93	-3.41	488	0.000	$[-28.96, -22.97]$	Y	-
B1-32–H-DAEDAL-1024	-31.16	-5.38	557	0.000	$[-34.27, -28.13]$	Y	-
B1-32–H-VAR	+0.30	-0.38	64	0.708	$[-0.83, 1.44]$	-	Y
B1-64–B1-128	-12.05	-2.27	357	0.000	$[-14.71, -9.25]$	Y	-
B1-64–B1-256	-15.39	-14.25	411	0.000	$[-18.27, -12.36]$	Y	-
B1-64–B0-256	-15.39	-14.63	403	0.000	$[-18.42, -12.59]$	Y	-
B1-64–H-PRED-95	-10.46	-1.14	334	0.000	$[-13.12, -7.88]$	Y	-
B1-64–H-DAEDAL-1024	-15.69	-3.11	401	0.000	$[-18.50, -12.96]$	Y	-
B1-64–H-VAR	+15.77	+1.90	384	0.000	$[12.96, 18.50]$	Y	-
B1-128–B1-256	-3.34	-11.98	250	0.006	$[-5.53, -1.14]$	Y	-

Table 15: J2 paired recomputation, part I (pairs 1–14 of 28). Δ_{can} and Δ_{strict} are arm-A-minus-arm-B accuracy differences in pp; m is the discordant-pair count and the interval is the item-level paired bootstrap on Δ_{can} . The complete inventory contains 23 pairs selected by the gate pre-registered in estimand wording, operationalized post hoc, and favorable in direction, with zero reversals among them.

pair	Δ_{can} (pp)	Δ_{strict} (pp)	m	p_{McN}	paired CI% (95%)	definite	rev
B1-128-B0-256	-3.34	-12.36	254	0.007	[-5.69,-1.06]	Y	-
B1-128-H-PRED-95	+1.59	+1.14	143	0.094	[-0.15,3.26]	-	-
B1-128-H-DAEDAL-1024	-3.64	-0.83	232	0.002	[-5.91,-1.44]	Y	-
B1-128-H-VAR	+27.82	+4.17	525	0.000	[24.94,30.71]	Y	-
B1-256-B0-256	+0.00	-0.38	24	1.000	[-0.76,0.76]	-	=
B1-256-H-PRED-95	+4.93	+13.12	287	0.000	[2.50,7.35]	Y	-
B1-256-H-DAEDAL-1024	-0.30	+11.14	162	0.814	[-2.27,1.59]	-	Y
B1-256-H-VAR	+31.16	+16.15	571	0.000	[28.05,34.42]	Y	-
B0-256-H-PRED-95	+4.93	+13.49	293	0.000	[2.35,7.35]	Y	-
B0-256-H-DAEDAL-1024	-0.30	+11.52	170	0.818	[-2.12,1.52]	-	Y
B0-256-H-VAR	+31.16	+16.53	565	0.000	[28.05,34.27]	Y	-
H-PRED-95-H-DAEDAL-1024	-5.23	-1.97	279	0.000	[-7.66,-2.81]	Y	-
H-PRED-95-H-VAR	+26.23	+3.03	490	0.000	[23.42,29.26]	Y	-
H-DAEDAL-1024-H-VAR	+31.46	+5.00	565	0.000	[28.28,34.57]	Y	-

Table 16: J2 paired recomputation, part II (pairs 15–28 of 28), with the same definitions as Table 15. Bold reversal marks identify the three raw canonical near-ties that flip sign; none passes the paired gate, pre-registered in estimand wording, operationalized post hoc, and favorable in direction, whose reversal count is zero.

D Same-bed parser-lineage audit

We re-score all 13 frozen cells (eight Bed 1 arms and five Bed 2 cells) with four deterministic text parsers. P1 is strict boxed-regex with a miss scored wrong. P2 applies that rule and then takes the last numeric span, re-implementing the canonical ruler with $6/2136 = 0.28\%$ drift, below the pre-registered 2% fidelity gate. P3 reproduces only the anchor-first regex tier of xFinder (Yu et al., 2024); P4 is an unconditional last-number baseline. The parser definitions and outputs are frozen with the artifact; Appendix A.1 lists the exact script and result paths.

Bed	Arm	P1 strict	P2 canonical	P3 xFinder-regex	P4 last- N
1	B1-32	5.16	45.79	44.88	45.79
1	B1-64	6.97	61.33	44.81	61.26
1	B1-128	9.40	73.39	39.50	73.01
1	B1-256	22.44	77.03	42.61	76.80
1	B0-256	22.59	76.95	43.29	76.72
1	H-DAEDAL-1024	10.77	77.10	41.62	76.88
1	H-PRED-95	7.88	71.87	40.71	71.80
1	H-VAR	5.53	45.49	44.50	45.49
2	B2-128	87.50	89.33	37.36	89.33
2	B2-96	83.71	84.83	40.73	85.11
2	B2-160	87.08	89.61	39.33	89.61
2	B2-64	69.38	71.07	54.78	71.35
2	B2-192	90.73	91.29	41.85	90.73

Table 17: Same-output accuracy (%) under four parser lineages. These are re-parses of frozen text, not new generations. P1/P2 are the operational ruler family; P3 is the regex tier, not the trainable LLM tier, of xFinder. The canonical re-implementation has six drift rows and passes its pre-registered fidelity gate. The low P1 strict cells on the short Bed 1 arms (for instance 5.16% on B1-32) measure how rarely those arms emit the boxed format, and must not be read against a chance-rate baseline: both beds use the free-form generative GSM8K split, so no option set, option count, or $1/K$ reference exists (§2, where the earlier “GSM8K-MC” alias and its above-chance wording are retracted). All four columns are therefore compared only against each other.

Parser	Definite B1	Definite B2	Reversals B1	Reversals B2
P1 strict	21	2	2	0
P2 canonical	23	3	0	0
P3 xFinder-regex	7	2	6	2
P4 last- N	23	3	0	0

Table 18: Pre-registered K2/K4 counts. A reversal is a relation definite under the named parser whose sign differs from the canonical reference; the table therefore corrects the earlier assumption that strict and canonical definite sets coincide.

K1 (parser sensitivity). Holm correction rejects 68 of 102 paired parser-disagreement tests, whereas the canonical-versus-last-number contrast rejects none of 34. Bed 1 rank order is identical for P2/P4 but not for the xFinder-regex tier, which changes the sign of 44 of 102 tested arm differences. Thus the canonical fallback acts almost like last-number extraction on these beds, while anchor-first parsing is structurally different.

K2 (nearest-neighbour decisiveness). Table 18 shows that the parser choice changes which relations are definite. The xFinder-regex tier depresses Bed 2 more sharply than Bed 1 and does not preserve the canonical ordering. This makes the same-bed comparison informative, while also showing that the manuscript’s operational-ruler conclusions do not automatically transfer one parser lineage away.

K3 (inter-trial re-extraction proxy). On shared-index complementary cells, canonical extracted answers agree on $253/356 = 71.1\%$ for the shorter pair and $325/356 = 91.3\%$ for the longer pair. This is an LLM-free determinism proxy only; it neither implements nor substitutes for Answer Regeneration.

K4 (gate survival). The strict parser has two definite Bed 1 sign reversals relative to canonical; the xFinder-regex tier has reversals on both beds, while the last-number baseline has none (Table 18). The same-bed evidence therefore establishes parser-lineage sensitivity rather than a parser-robust zero-reversal law. The fixed-128-versus-160 Bed 2 relation remains non-definite under every tested parser, so it does not acquire a decisive pattern label.

Pre-registered falsification criterion. If a subsequent panel still records “nearest neighbours never compared on the same bed” after this appendix, then the regex tier was insufficient; the nearest-neighbour goal moves to admitted future work rather than being claimed closed.

Boundary and limitations. This is the xFinder regex tier only. The LLM tier and Answer Regeneration are reservation-gated; K3 is an LLM-free proxy. No parser is evaluated beyond the two frozen beds, and P2’s near-equivalence to P4 cannot be attributed to its boxed-first step alone. This appendix is therefore partial, not full, closure of the cited-but-not-compared gap.

E Deviations from the preregistration (full list)

The preregistration (notes/prereg_crossbed_judge_normalization_r115.md) fixed the rules in advance; this refine discloses every place the executed analysis or the revised draft departs from it, with the directional effect of each deviation.

- J3 estimand. The preregistration specifies a paired McNemar for the B2-128–B2-160 contrast with an $m \geq 25$ disclosure threshold. An earlier draft executed an unpaired two-proportion interval on the full pool; this draft returns to the registered paired test on the common 356-item support ($m=40$, §5.4). Direction: brings the claim back to the registered estimator.

- J3 population. An earlier draft compared the full 712-pool B2-128 with the 356-item over-half, producing the spurious -0.08 pp tie. Retracted in favor of the common-support contrast. Direction: acknowledges the non-common selected populations that the preregistration’s J1 rule already blocks.
- J3 verdict estimand (revision). An earlier draft labeled J3 survives “under the joint/strict estimand only,” letting the auxiliary (fallback-stripped) estimand carry the survive verdict while the primary conditional estimand ($|\Delta_{\text{true}}| = +1.02$ pp > 0.5 pp) exceeded the explicitly post-hoc tolerance. The preregistration gives adjudication to the primary estimand only, so this revision re-decides J3 on the primary as not_identified from the shipped summaries—the data cannot decide (power-limited: the post-hoc 0.5 pp primary MDA80 is not identified; the canonical/strict secondary-ruler MDA80 values are 4.719/4.977 pp, the primary conditional CI implies only a retrospective post-hoc resolution of ≈ 4.57 pp, and the item-paired 95% CI on Δ_{true} contains zero, §5.4) and reports the exact joint tie as a sensitivity finding; failing to reject the null is not proof of no effect. Direction: more faithful to the registered primary/secondary split; changes the J3 label from survives to not_identified.
- J3 u resolution. Enumerating the over-half items resolves $u=5$ (an earlier $u \in \{4, 5\}$ bracket), pinning $\Delta_{\text{true}} = +1.018$ pp. Direction: no change to the primary-estimand verdict; converts a “verifiable” bound into a verified point value.
- J2 “definite” operationalization. The preregistration fixes the word “definite” but not its criterion. This draft operationalizes it (95% CI excludes 0; $|\Delta| \gtrsim 3.7$ pp) after inspecting the data, in the direction that preserves the claim, and discloses the rule-dependence in §5.3. Direction: data-determined and directionally favorable; not part of the pre-registered threshold set.
- J1 capability reading. Prereg verdict J1, branch 2 reads an interval excluding zero as a nonzero model-capability residual. This draft instead treats the contrast as descriptive because the selected populations differ. Direction: more conservative, declared post-hoc, not presented as preregistered.
- J5 synthesis. The registered rule is a conjunctive binary verdict (all of J1/J2/J3 containing zero $\Rightarrow$ comparable). We report an order/pattern-vs-level hierarchy instead, with explicit primary-arm decision rules for J1/J2/J3 (§4, Primary-arm decision rules). Direction: a graded statement that changes the single negative verdict; declared post hoc; it was pre-registered only in estimand wording.
- Headline statistic. The preregistration lists the 48-item 91.7% reference figure; the head number here is the J1 arm’s item-level 88.6% ($n=604$), with 91.7% kept as the secondary reference subset. Direction: no conclusion change.
- J2 arm inventory. The preregistration and the frozen table assert nine Bed 1 arms / 36 pairs; only eight arms (28 pairs) are reproducible from the shipped per-arm records, and the ninth arm is not identified. We enumerate the reproducible 28 and disclose the inventory gap rather than asserting the 36.
- J2 definiteness gate. Independently of the inventory gap, the lead raw rule reads 3/28, while the favorable paired canonical-definite gate—pre-registered in estimand wording and operationalized post hoc—reads $F_{\text{strip}} = 0/23$; the strict- and either-definite sensitivity gates each read 2 (§5.3). The two looser-gate reversals involve H-DAEDAL-1024. Direction: the paired gate is claim-preserving and must travel with all three alternative readings.
- J2 zero-event bounds withdrawn (this revision). Earlier drafts attached one-sided 95% upper bounds of 12.2%, 11.3%, and 77.6% to the zero-reversal counts 0/23, 0/25, and 0/2 in Table 6. Each was computed as $1 - 0.05^{1/n}$, i.e. by treating the eligible pairs as independent Bernoulli trials. That assumption is false here: every Bed 1 pair is scored on the same 1,319 problem IDs, and the merged 0/25 set contains the same 23 pairs, so the bounds overstate confidence and the 11.3% cell was not even independent of the 12.2% cell it sat beside. All three are withdrawn as inferential statements and replaced by the item-paired strata; the arithmetic remains reproducible from n alone for anyone auditing the retraction. Direction: strictly weakens what the zero-reversal readings are allowed to support.

F Interpretation checklist

The following restrictions are part of the result, rather than optional qualifiers; Tables 19 and 20 keep each restriction beside the result it qualifies.

Result	Required interpretation
J1	Conditions on post-outcome (post-generation) behavior and is a descriptive selected-population contrast, not a capability ranking or a causal effect.
Decomposition	The additive share is not identified: its legal range is [59.11%, 95.87%]; the anchor-free multiplicative log share is 82.58%. The old 58.4% residual bookkeeping is not an identified decomposition.
J2	The raw point-estimate rule leads with 3/28; the directionally favorable paired gate, pre-registered in estimand wording and operationalized post hoc, reads 0/23; each looser definite gate reads 2 (§C); this does not identify a population reversal probability of zero.

Table 19: Interpretation checklist, part I: level, decomposition, and order. These restrictions travel with the corresponding results.

Result	Required interpretation
J3	not_identified from the shipped summaries on the preregistered primary ($P(\text{correct} \mid \text{fnt})$) estimand—the data cannot decide: $+1.02 \text{ pp} > 0.5 \text{ pp}$ under the explicitly post-hoc tolerance, but the item-paired 95% CI $[-2.23, +4.17] \text{ pp}$ contains zero. Primary-ruler MDA80 is not identified; canonical/strict secondary-ruler MDA80 values are 4.719/4.977 pp, and the primary conditional CI gives only a retrospective, post-hoc $\approx 4.57 \text{ pp}$ resolution. The exact joint tie holds only under the auxiliary strict estimand, and neither interval is narrow enough to establish equivalence within the post-hoc 0.5 pp margin.
Final synthesis	The internal six-seat panel on the refine-4 frozen SHA returned overall 6 (marginal_accept), soundness 3, and contribution 2; the shipped external gpt-5.6-sol review of the same SHA returned overall 3. The manuscript is positioned against the external grade, not the internal panel.
Reuse	The per-arm sensitivity table reuses the frozen Bed 1 per-arm strict-decomposition assets and adds no independent evaluation bed.

Table 20: Interpretation checklist, part II: pattern, synthesis, and reuse. Together with Table 19, this changes presentation, not conclusions.